Artificial Intelligence and Machine Learning(6CS012)

Part I – Question and Answer (QnA)

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Long Questions

2. You are a Machine Learning Engineer at a rapidly growing e-commerce company responsible for deploying and maintaining ML systems in production.

Machine Learning (ML) systems face various challenges from technical, operational, business domains, organizational, and legal/ethical. The following are three real-world challenges encountered in ML systems:

a. Data Quality and Availability

Poor or biased data is fundamental failure mode. If training data is incomplete, lacks volume or mislabeled, the model will be garbage-in/garbage-out. Garbage doesn't often become apparent until models fail to converge or make unreliable predictions late in development.

Consequences:

- Models underperform.
- Predictions can be wrong.
- Wasted resources and computational power.

Solutions:

- Data validation.
- Fallback Data & Redundancy.
- Synthetic Data & Data Augmentation.

Trade-offs:

- Data validation is cheap and effective for obvious errors but cannot catch semantic issues (e.g. mislabeled product categories).
- Synthetic data may introduce artificial bias if not carefully validated.

b. Model Bias and Fairness

ML models can inherit and amplify social biases present in data. If training data over-represents one group, models will favor that group which is unfair to others.

Consequences:

- Biased models can cause legal and reputational damage. For example: discriminatory lending, hiring, or content moderation.

Solutions:

- Diverse training data.
- Conduct fairness testing.
- Establish governance.

Trade-offs:

- Imposing fairness constraints often reduces overall accuracy.

c. Privacy and Data Governance

Privacy laws mandate strict controls. Moreover, consumer trust hinges on data protection. Without controls, it's easy to inadvertently leak personal data (e.g. unsecured logs, or unsecured APIs).

Consequences:

- Data breaches and leaks
- Regulatory fines
- Customer distrust

Solutions:

- Access controls.
- Consent & governance workflows.
- Model encryption.

Trade-offs:

- Data privacy guarantees privacy but can severely reduce accuracy.
- Governance systems (catalogues, audits) improve oversight but can become stale if not regularly updated.

Cross-Functional Collaboration:

Clear communication between various roles is vital for building reliable ML systems as it requires coordination between data scientists, data engineers, ML engineers, legal teams and product managers.

- Product managers set priorities that align with customer needs.
- Legal team reviews privacy impact and regulatory compliance.
- Data scientists might detect bias in a model.
- Data engineers and ML engineers support to trace the bias and update the pipeline.

Modern Tools:

- Automation tools like Azure ML can orchestrate end-to-end workflows which enable reproducibility and easy integration of tests.
- SaaS platforms like Prometheus automate performance and fairness monitoring.
- LLMs like Claude Code and Cursor can accelerate code and report writing.

Short Questions

Overfitting

2. Overfitting is a common challenge in deep learning, especially when models have high capacity.

When a deep learning model learns training data, including noise and irrelevant patterns—too well, it is said to be overfitting and performs poorly on new data.

Following two techniques are used to reduce overfitting:

Early Stopping:

It is a regularization technique used to stop a model's training when its performance on a validation set stops improving, even if training loss is still decreasing. By stopping before the model learns noise, it improves generalization.

A model's training is monitored using loss/accuracy. When performance stops improving, training halts. This avoids excess fitting and reduces computation.

Example: To make sure the model applies to new patients, training in a medical image classification is stopped when validation accuracy reaches a plateau.

Data Augmentation:

It is a technique that uses pre-existing data to generate new data samples that can improve model optimization and generalization instead of memorizing specific examples.

It increases data diversity, which is crucial for deep learning by applying transformations like rotation, flipping, or color adjustments to images without the need for new data collection.

Example: Images are augmented with brightness changes and weather effects in Autonomous Driving Systems, so a vehicle can identify pedestrians, traffic signs and signals better.

Neural Network Architecture

1. Explain the fundamental differences between a Convolutional Neural Network (CNN) and a Recurrent Neural Network (RNN).

Convolutional Neural Networks (CNNs) are feedforward networks specialized for spatial data like images and videos.

Recurrent Neural Networks (RNNs) are sequential networks that process data one step at a time carrying a hidden state forward.

Input structure and data type: